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Considering the Hydrologic Cycle: Eighty Years of Graphical Grappling

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Abstract: In 1934, the Natural Resources Board of the United States of America published the first visually descriptive hydrologic cycle diagram. Like water itself, this simple graphic has evolved in some ways and remained stagnant in others in the past eight decades. Multiple edits have been made, graphics have become more realistic, and many agencies and organisations have developed their own diagrams. Yet the majority of hydrologic cycle diagrams continue to ignore or understate the role of humans in the hydrologic system and the vast diversity of watersheds.

This review explores the thinking of over one hundred scholars grappling with the interactions between society and water systems. Geographic, environmental, anthropological, political, and economic perspectives are all considered. The inconsistencies and tensions found between and within articles highlight the complexity of water and our ongoing struggle to truly understand it. They also indicate that it is long past time for a more nuanced diagram incorporating the myriad way(s) water moves and flows to take the place of the "classic" hydrologic cycle.

Introduction

Earth and biological sciences conceptualise any number of processes as cycles. Children are taught from an early age various organisms' "life cycles", even as individual organism's lives are chronologically linear from a biological standpoint. Elements, too, are understood and taught as cycles: Literature frequently refers to multiple cycles such as the carbon (Russi et al., 2013), nutrient (ibid), ozone (McLaughlin et al., 2007), nitrogen (Vitousek et al., 1997), and environmental (McIntyre, 1972).

Water is no exception. The "water cycle" or "hydrologic (hydro) cycle" is practically a household name, especially for families with children in primary school. The basic pattern of precipitation, evaporation, and condensation forms the basis of how most people understand and view water.

Yet these basic biophysical processes that are captured as cycles do not exist in simple isolation. The cycles and systems we have individually categorised influence each other (Linton and Budds, 2013; McLaughlin et al., 2007). Humans and other animals influence and help to shape these cycles in addition to being shaped themselves, in ways beyond those generally represented.

This literature review will consider the ways the hydro cycle is represented and understood. We will begin with an overview of the "classic" hydro cycle, exploring the model's history. We will then turn to the many issues *not* included in this classic cycle, most notably the role of human society and usage. Emerging theories about how water flows could be more fully understood and incorporated will be addressed. The review will conclude with some thoughts about the next steps for scholarship on the hydro cycle, calling for inter- and trans-disciplinary conversation about re-envisioning the way water flows are taught and depicted.

The "Classic" Hydro Cycle

History

Jamie Linton's *What is Water?* (2010) reviews the history of the hydrologic cycle, providing a helpful distinction between the origins of hydrology as a scientific field and the creation of the diagrammatic depiction of the hydro cycle. The hydro cycle diagram as we think of it today is a modern creation less than a hundred years old, but humans have been considering water for millennia, often in a cyclical fashion. The Hebrew Bible contains an allusion to water: "All the rivers run into the sea, yet the sea is never full; unto the place from whence

the rivers come thither they return again" (Ecclesiastes 1:7, quoted in Leopold, 1960). Many Greek and Roman philosophers considered water flows; Hippon or Thales are thought to have developed a seawater filtration theory, arguing that seawater filters up through the land and percolates as surface water back to the sea (Brutsaert, 2012). Attention to water movements was not limited to just scientists and philosophers. Poetry written by Hesiod in the eighth century B.C.E. includes evidence of considering these movements (Brutsaert, 2012).

Pierre Perrault can be considered the first "pure" hydrologist, as his 1674 book *On the Origin of Springs* sought to describe and begin quantifying water in its different forms. Russian hydrologist E. Ya. Brickner continued this work in 1905 in his attempt at quantifying global water resources (UNESCO, 1971). Around this time, hydrology became more explicitly defined as an academic discipline. In 1899, W. M. Davis conceptualised the "geographic cycle", helping to establish and set the parameters for the science of geography. A 1931 paper by Robert E. Horton called, "The Field, Scope and Status of the Science of Hydrology", followed suit for hydrology. The paper included the first notable diagram depicting water movements and an argument for separating the natural and social aspects of water, sparking discourse around creating a diagrammatic representation of biophysical water flows. Diagrams varied extensively for the first decade, as hydrologists considered various potential depictions. Adams considered the water diagrams of early Greek philosophers in his 1938 book. Eventually, Thorndike Saville coined the term "hydrologic cycle" to describe the diagrammatic depiction that we will refer to in this paper as the "classic" hydro cycle (Linton, 2010). The National Resources Board published early versions of the hydro cycle diagram in 1934 (Linton, 2008) along with Mienzer in 1942.

Today's hydro cycle diagrams look much different than Horton's first cycle (**Figure 1)** in 1931, but surprisingly similar to the one published by the National Resources Board in 1934.

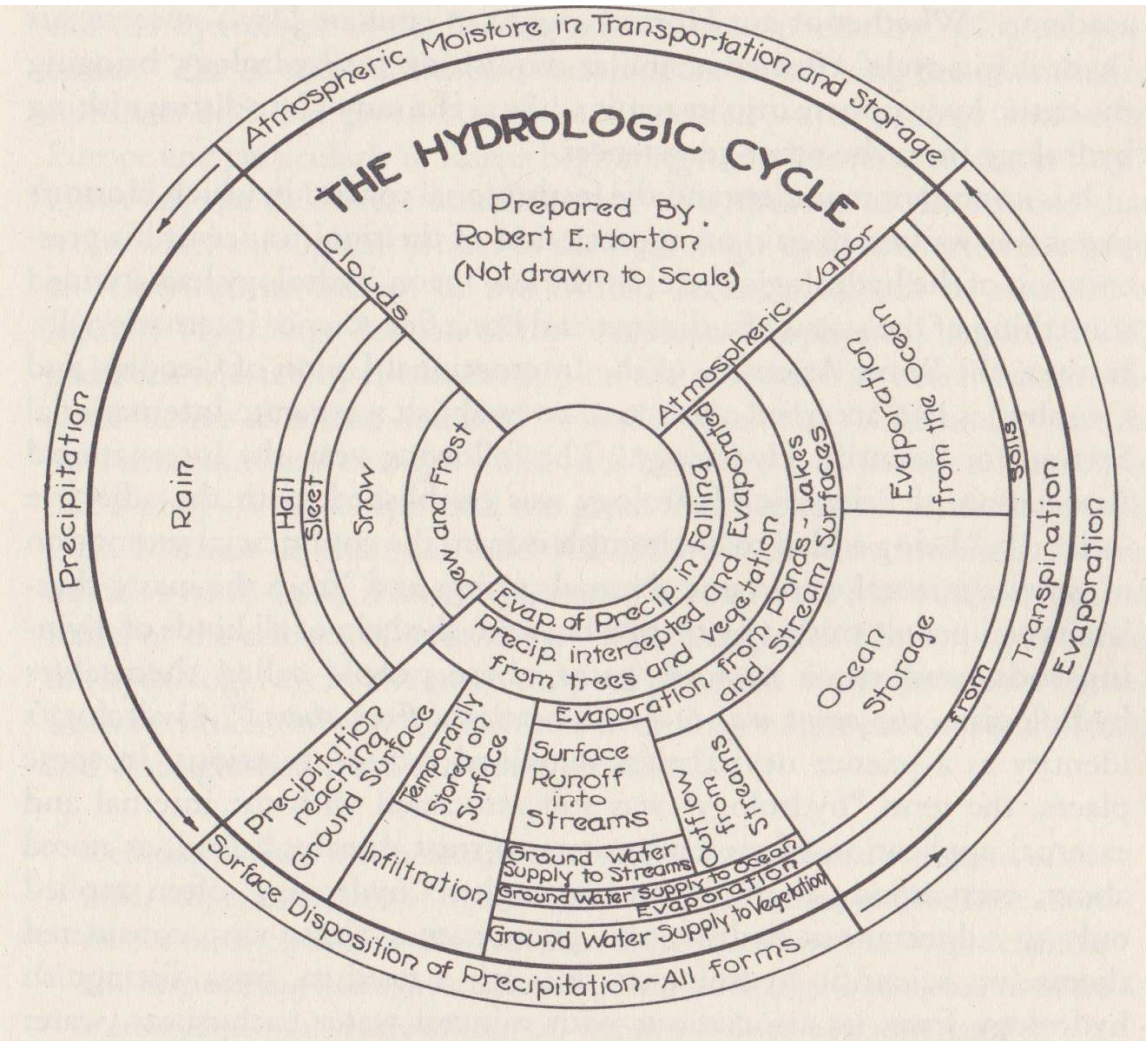
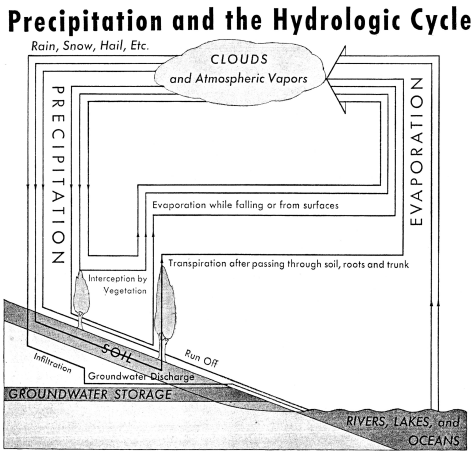


Figure 1. "The Hydrologic Cycle" from Robert Horton (1931)

Changes have mostly been graphical rather than conceptual: The diagrams found in an average primary school textbook look more realistic, but they continue to have arrows representing unidirectional flows of water through almost entirely biophysical processes. Compare **Figure 2** ("Precipitation and the hydrologic cycle", National Resources Board, 1934: 262; from Linton, 2008: 638) with **Figure 3**, the current depiction of the hydro cycle by the United States Geological Survey (USGS, 2013). The graphics are improved aesthetically and more processes are shown in **Figure 3**, but the two look surprisingly similar conceptually, given that nearly eighty years of modern science separate them. As such, the USGS' current hydro cycle can be considered a "classic" hydro cycle.



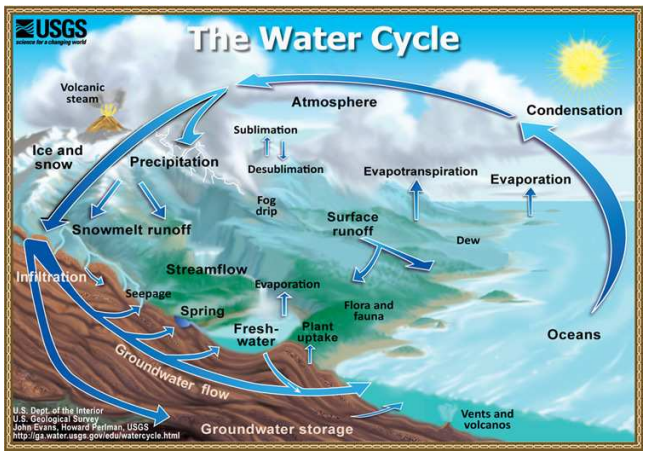


Figure 2. (left) National Resource Board, 1934 (Linton, 2008)**.** **Figure 3. (right)** USGS, 2013.

Education and Public Discourse

Since the 1934 publication of the U.S. National Resources Board's hydro cycle diagram, various versions of the classic hydro cycle are used to teach hydrology. There has been relatively little research on students' knowledge and perspectives of the hydro cycle (Shepardson et al., 2009), though it is taught in most classrooms as part of science curriculums.

A basic understanding of the classic cycle is necessary for what Cockerill refers to as "water literacy" (2010: 151). Cockerill posits that not understanding the basics of hydrology (e.g., believing that the planet could literally run out of water) decreases people's willingness and capacity to take action or change habits around water use. Cockerill ran a programme in North Carolina "translating" hydrology and scientific knowledge to make it accessible for a general audience, reinforcing the classic hydro cycle. Taiwo et al. (1999) explore how the hydro cycle is understood by schoolchildren in Botswana. They argued that schooling "positively influenced" children's perceptions of the hydro cycle while the "'untutored ideas the children brought to school'" (e.g., "clouds are made by gods") negatively influenced their knowledge of water.

Not all educators seek to reinforce so strongly the classic hydro cycle as such. Davies and Seimears (2008) suggest that "unpacking" the multiple components of water (chemistry, biological function, societal uses, etc.) is necessary for teaching. Students, teachers, and groups can then pull ideas and issues together. Similarly, Eisen et al. use water as a case study for interdisciplinary teaching; their 2009 paper provides a guide for two water modules that combine the literature, art, philosophy, and science of water.

In many ways, pedagogy and educational practice are reflections of broader public discourse. Public participation around water-related discourses influences their knowledge of and interactions with water (Fosen, 2012). For example, Moran (2008) discusses how people's engagement with composting toilets and greywater systems in their homes can motivate and catalyse policy around sustainability, while hidden, unacknowledged septic systems in the ground do not help improve awareness. At the same time, Stenekes et al. (2006) urge against supporting rhetoric that blames "public ignorance" and "cultural bias" for the failure of programmes like water recycling, believing that this produces and reinforces a dichotomy between lay and expert opinions around water issues.

Gaps in the Classic Hydro Cycle

As Fosen (2012), Moran (2008), and Stenekes et al. (2006) demonstrate, public discourse does not stay stagnant over time. Nor does academic discourse. This section seeks to outline some of the many issues about water that academics explore and are not adequately conveyed by the classic hydro cycle. Barnes (2001) reminds us that the dominant theories leading academic disciplines change over time. The classic hydro cycle was created in a time when epistemological theorising was dominant. However, Barnes suggests that hermeneutic theorising is gaining prominence in economic geography, focusing on interpretive and reflexive thinking and work. As underpinning theories and assumptions change in a discipline, so too should that discipline's primary teaching and communication tools. Today's knowledge is different than the knowledge dominant as recently as the 2000's, and very different from that of the 1930s'. Beck's 1984 piece written as a, "Topic of Public Interest: Water Quality", reflects on how post-war management impacted the collection and availability of empirical data, and thus academics' ability to pay more attention to individual behaviour, pollution, and the like. Beck's work serves as another reminder that we notice and know more now than we did when the hydro cycle was first conceptualised and created; because of this, it cannot be expected to retain its power or accurately reflect what we know about water today.

This is not to say that we know everything, or that a "perfect" model could be created. Beven's 2006 writing, "Searching for the Holy Grail of scientific hydrology", highlights just how difficult perfect modeling is. Beven acknowledges that the non-linearity, fluxes, and storages of water are hard to know entirely, and states that, "The closure problem [the boundary fluxes of mass, energy, and momentum in a watershed] is a scientific Holy Grail:

worth searching for even if a general solution might ultimate prove impossible to find," (609). Similarly, a fully nuanced diagrammatic representation of water flows that can be used to teach and learn is something to strive toward, even as we know it cannot be truly achieved.

Local Climate

A great deal of the power and usefulness of the classic hydro cycle is its simplicity and apparent intuitiveness. But, the standard hydro cycle depiction is really only accurate for a very specific climate. Most of the planet is not the cross-section of temperate green land with one river and a visible ocean depicted in the classic hydro cycle, yet people are being taught this hydro cycle, even when irrelevant to their locales. While examining students' understanding of hydrology, Shepardson et al. (2009) found that, "[m]idwest students often portrayed the hydrologic cycle in the context of mountain or coastal landscapes that are common in textbooks but that are not representative of the environments where students live and where many of these students might apply their understanding of environmental systems as adults." (2009: 1447).

Scale and Unit

But if the classic hydro cycle only describes a particular landscape and does not accurately reflect the bulk of water flows on Earth, which unit should be considered? Hydrologists and other scholars deal with water issues at virtually every scale, but the most common unit for water management is almost certainly the river basin (used by Richter et al., 2003, among many others). Most depictions of the classic hydro cycle are centered on a single river. Keys et al. (2012) focus on rivers because they "connect upstream and downstream ecosystems within watersheds," (2012: 733). Yet rivers are not the only way water moves in the world, and the river basin is not necessarily the most appropriate unit to be examining as only a tiny portion of the Earth's water resources are contained in rivers. Warner et al. (2008) argue that the decision to focus on the river basin as a unit for management and science is as much a political one as a natural one.

Beyond the river basin, Kalinin (1971) built on fairly early conceptions of the hydro cycle and hydrology to develop a concept of global hydrology. More recently, Kotwicki (2009) suggests considering both a hydro-tectonic water cycle (conceptualized as the loss and gain of water from surface into and out of the Earth's mantle) and an Earth-outer space water cycle (a long timescale process, but one that may involve net water gain for the planet over time). Kotwicki also acknowledges the temporal element of unit, highlighting the importance

of creating a hydro cycle that is accurate on a geological time scale. Howarth's 1986 article explores the atmosphere's role in the hydro cycle on a hemispheric level. Growing in prevalence are concepts such as the "precipitationshed", seeking to use those components of ecosystems and water that impact each other as a combined unit, regardless of their spatial placement (Keys et al., 2012).

Human Influence

Concepts like the "precipitationshed" recognise that actions in one area impact water availability, quality, and flows in other areas. Local actions have global impacts; global trends affect local issues. "Human-induced changes to the water system are now global in extent, yet we lack an adequate understanding of how the overall system works" (Vörösmarty et al., 2004: 513). Fonstand's (2013) introduction to the "Geographies of Water" special issue of the *Annals of the Association of American Geographers* points out that many of these human-induced changes are not new, but neither are they fully known. It is clear we need to better understand and deal with the connections between humanity, hydrologic flows, and ecosystems.

Perhaps the thing most obviously missing from the classic hydro cycle are humans: The vast majority of diagrams do not include a single individual or societal influence, but rather appear to be animal- and human-free landscapes. Yet humans use, disrupt, redirect, and recycle water flows in a multitude of ways. Issues around human influences in and from the hydro cycle that are not included in the classic hydro cycle include: human demand and population growth (Merrett, 2004; Roelke, Spatharis, and Mitrovic, 2012); economics (Merrett, 2004; Swyngedouw, 2006), including understanding water as an economic resource (Linton, 2006), the role of industry (Davies and Seimears, 2008; Peters and Meybeck, 2000), water's use in power generation (Davies and Seimears, 2008), water markets (Embid, 2010), and human ownership of water (Swyngedouw, 2009); hydro-social interactions (Linton and Budds, 2013); hydro-social modernisation (Swyngedouw, 2007), including the hydraulic mission (Swyngedouw, 2007), dams (Rudolph et al., 2013; Graf, 2001), pollution (Jain and Singh, 2010), recreation (Davies and Seimears, 2008), recycling (Jain and Singh, 2010), human environmental footprints (Hoekstra, 2011; Leu et al., 2008), urbanisation (Masjuan et al., 2008; Swyngedouw, 2004; Swyngedouw et al., 2002), and household septic systems (Moran, 2008); social institutions such as water policies and law (Buyukcangaz and Korukcu, 2007; Embid, 2010; MacDonnell and Grigg, 2007), conceptions of water as a human right (Leb, 2012), peacebuilding practices (Weinthal et al., 2011), and extreme event management

(Jain and Singh, 2010; Glennon, 2009); land use (Cohen, 2012; Kagawa et al., 2009) including agriculture, food, and irrigation concerns (Davies and Seimears, 2008; Keys et al., 2012; Rudolph et al., 2013; Weinthal et al., 2011); politics around nationalism and identity formation (Desbiens, 2004), borders and transboundary issues (Leb, 2012), power and governance (Linton, 2011; Zeitoun and Warner 2005; Swyngedouw, 2002; Swyngedouw, 2006); spiritual and religious uses (Tuan, 1968); and virtual/embedded water flows (Allan, 2003; Hoekstra, 2003).

The above list is long, a reflection of just how many ways humans impact – and are impacted by – water. Any nuanced depiction of water flows in our world must engage questions of human-environment interaction.

Biophysical Processes

It is not only sociopolitical issues that most hydro cycle diagrams miss. Multiple biophysical processes are not conveyed well by the classic, simplistic model. Leopold's 1960 hydrology primer emphasises the importance of realising that most water vapour comes from oceans and evaporation from the land, rather than lakes and rivers, where it's generally shown as taking place on classic hydro cycle diagrams. Nor does the classic hydro cycle adequately represent the interconnection of local cycles (Roelke et al., 2012); the "non-uniform distribution of water resources all over the world" (Buyukcangaz and Korukcu, 2007: 710); bioprecipitation, the idea that microbial ice nucleators may play a significant role in the formation of rain and snow (Cohen, 2012); the multiplicity of ecosystems, vegetation, and plants (Elmore et al., 2003; Kagawa et al., 2009); floods, droughts, and other water hazards and extreme events (White 1945; Glennon, 2009); or the reality that change, rather than equilibrium, can be seen as nature's default (Graf, 2001).

Climate Change

Many of the issues above, both sociological and biophysical, contribute to and are impacted by climate change. The role of climate change in the hydro cycle is considered by authors like Roelke, Spatharis, and Mitrovic (2012); Terray et al. (2012); and Arreguín-Cortés and López-Pérez (2013). Global temperature rises affect evaporation and transpiration rates, potentially creating losses in available water quantities for localities (Arreguín-Cortés and López-Pérez, 2013). In spite of the range of impacts and the need for response, the static nature of the classic hydro cycle means that climate change is not incorporated in the basic model.

The Hydro-Illogical Cycle?

The classic hydro cycle is missing so many issues that strongly influence water movements that some authors refer to it as the "hydro-illogical" cycle (Wilhite, 2011). The term is also used to refer to human perceptions of and reactions to drought and concerns that the cycle is somehow "broken" (Glennon, 2009).

While the classic hydro cycle has been a powerful tool for teaching and communication, its gaps are many. Phrases like the "hydro-illogical" cycle show that scholars now consider the classic model inadequate, and perhaps even harmful. In this next section, we consider some of the burgeoning theories around ways to enhance or replace the basic model.

Moving toward a More Holistic Hydro Model

Society and the Environment

Science has a history of separating society and human endeavour from the environment. Perrault's 1674 *On the Origins of Springs* suggests looking at water in its different forms as a separate issue apart from water's human use. Horton's 1931 article presents hydrology and cycle diagrams as a solely scientific process, rather than the product of an author's ideas.

Jamie Linton and Jessica Budds (2013) explore the separation of society and human creation from environmental processes, the social-environmental dichotomy, which has drawn critique from multiple disciplines and authors. Few people would agree that the dichotomy is absolute; humans influence water and water influences human society. As the conference announcement for a session on "Mountains to Sea: Human Interaction with the Hydrologic Cycle" said:

> "The hydrologic cycle is the principal, dynamic driver for many of the world's natural physical processes. Water in its abundance or absence has a profound effect on human survival and societal development. In turn, human activity may intentionally, or unintentionally, alter the pathways and quality of water in the hydrosphere, biosphere, and atmosphere. Human intervention in the hydrologic cycle may produce beneficial or disastrous results" (*Canadian Water Resources Journal*, 1998).

There is a "complex web of interaction" and a great many "feedbacks" in human-environmental relations (Harden, 2012). Castree (2002) sees a society-environment nexus rather than dichotomy, shaped by a dialectical synthesis between humans and their

environment. The relational dialectic between water and society is discussed by Linton and Budds (2013), Linton (2010) and Loftus (2010).

In opposition to Horton's vision of the hydro cycle as unbiased by scholars, Budds (2009) points to the wealth of literature arguing that physical assessments are not neutral and that studies are shaped by users' understandings. In "Privatizing Water, Producing Scarcity: The Yorkshire Drought of 1995", Karen Bakker challenges conventional interpretations of the Drought, arguing that drought can be understood as the *production* of scarcity through the combination of meteorological modeling, demand forecasting, and corporate restructuring and regulation. The work of Budds, Bakker, and similar authors argues that the classic hydro cycle is the product of human framing rather than external, static, biophysical fact.

Because of this, Budds (2009) argues that we need to consider both socio-political factors as well as geoclimatic ones in analysing waterscapes. In 1969, a UNESCO report noted that, thus far, "our attempts at 'mastery of the environment' have been mere short-sighted tinkerings with the landscape," (Nace, 1969: 9-10). When "[e]veryone lives downstream of the effects of some human activity," (Peters and Meybeck, 2000: 185), we need to understand "the ability and limits of freshwater ecosystems to respond to human-generated pressures" in the midst of "altered hydrological regimes," (Naiman and Turner 2000: 958). In a 2012 article, Xifeng et al. advocate for a dualistic water cycle model that incorporates both biophysical and sociopolitical issues in China's Hun River Basin in order to better understand water use for management. Shiva (2008) discusses the importance of considering cultural attitudes around water use when making and adapting policy and science.

Reframing the Hydro Cycle

Alternatives and additions to the classic hydro cycle are both qualitative and quantitative in nature. Some seek to know water's actual movements more precisely but continue to focus primarily on water; others seek to place water in the midst of many other processes or focus on the multidirectionality of water flows.

On the quantitative end, home hydrologists perform "water balance" analyses. Water balances, or water budgets (Kalinin, 1971) can be seen as a type of hydro cycle focused on the scientific measurement of water volumes in a location and the flows of water in and out of a system (Thornhwaite and Mather, 1995). The approach is perhaps most commonly used in soil; Zieri (2001) shows how a water balance model can be used to understand soil

moisture and help predict drought conditions. The water balance approach is also used to examine things like glacier characteristics (Matsumoto et al., 2004), the impact of tree cover (Joffre and Rambal, 1993), and climate change impacts (Porporato, Daly, and Rodriguez-Iturbe, 2004). Kotwicki (2009) expands the typical scale of the water balance approach dramatically, exploring the water balance of Earth.

Water balances are generally quantitative and scientific approaches. Nakayama and Shankman (2013) use the idea of the water *im*balance to frame the impact large anthropogenic projects can have on water balances. Using case studies like the Three Gorges Dam and the South-North Water Transfer Project in China, Nakayama and Shankman propose creating an eco-hydrologic model to determine the socio-economic and environmental consequences of human activities.

Considering directionality, Wang et al. (2009: 499) speak of the need for "nonlinear correction to the hydrological cycle" while exploring natural and artificial drivers of changes in the hydrologic system of the Yellow River Basin. Nace (1969) refers to the water "wheel"; Peters and Meybeck (2000: 185) speak of water "pathways". Davies and Seimears (2008) have created a diagram of some of the various components of water that can be used in education. Their diagram of how these issues are connected contains many multidirectional arrows and suggests a "hydro web", rather than a hydro cycle.

Seeking to incorporate more than biophysical processes like gravity and cohesion in our analyses of how water flows in the world, several scholars engage with the socio-hydrological cycle (Swyngedouw, 2006), hydrosocial cycle (Linton, Budds, and McDonnell, 2013), and socio-cycle of water use and management (Turton, 1999). These versions of the cycle include "a flow not only of H_2O, but also one that is saturated with all manner of power relations" (Swyngedouw, 2006: 15). Swyngedouw uses the concept of "water metabolisms" to capture the multiplicity of meanings and characteristics water has through biophysical properties, cultural symbolism, and socioeconomic concerns (Swyngedouw, 2009). Linton and Budds (2013) expand on the social dimension of water by addressing both social and political influences. They emphasise the two-way relationship between society and water stating that the hydrosocial cycle is a, "socio-natural process by which water and society make and remake each other over space and time."

Conclusions and Next Steps

This literature review is intended to serve as a broad overview of issues surrounding the hydro cycle today, and is thus limited in its scope of detail. A full literature review could – and should – be undertaken on each of the topics mentioned above.

Given the many gaps that have been identified in the "classic" hydro cycle in addition to the theoretical frameworks that are coming to fruition around more complex ways to view water, human thinking in regard to water has moved forward since the basic diagram was created in the 1930s. The diagram has not evolved as fully as our thinking. This resulting gap provides a clear action step for further work. It is time to re-envision the classic hydro cycle. The authors of this literature review have convened a working group to do just that and are currently working on a follow-up piece to this review introducing one possible alternative. We seek feedback on this forthcoming suggestion and welcome additional ideas.

Additionally, it is clear that continued and enhanced collaboration around water and hydro flows is necessary. This collaboration needs to be done in inter- and trans-disciplinary settings involving a wide range of water stakeholders such as hydrologists, anthropologists, political scientists and economists, and moving beyond academia to include policymakers, educators, water practitioners, and water users. Beyond the complexities of bridging academic disciplines, further study should also be done among people who view water in fundamentally different ways. Exploration should happen not only across the well-known private vs. public and human right vs. economic resource debates, but also between those who view water as a fundamental element and those who see it as a social construction. Multiple perspectives on water, some of which seem to be mutually exclusive, must nonetheless be considered together if we are to arrive at a more nuanced understanding of the hydro cycle and water itself.

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